Project Report

Design Approach

The project uses stm32f103c8 as the MCU where MATLAB generated time series data is used as voltage and current reads, mimicking real world scenarios. The data was passed into "File Generator" module in proteus which generated alternating unstable voltage/current behavior spanning from over-voltage, under-voltage and over-current. This data was directly fed-into input pins of stm32 MCU.

The firmware enables both on-device settings configuration and computer based settings configuration. Where MODE and ADJUST button can be used for on-device configuration of settings like over-voltage limit, under-voltage limit, over current limit etc. The same configurations can be achieved through serial monitor text commands on PC by using CP2102 or CH340 USB-Serial adapters.

Source Code Writing and Build Tool Usage

This project is implemented and simulated on proteus 9 professional. The project files were written on boilerplate codes generated by STM32CubeMX with some additional C source files. The project is built using "GNU tools for stm32 14.3" (arm-none-eabi-gcc). Running the building batch file "build.bat", will build the entire project and generate .elf, .hex and .bin files.

Result

The final project is a realistic voltage/current surge protection device that is tested against real world voltage/current surge behaviors, which successfully detects over-voltage, under-voltage, over-current in the incoming source line and trips accordingly. But there are limitations on unexpected scenarios for which the auto-restore delay requires redesigning of logic and optimization.

In the proteus environment, the operation of toggle switch for on-device configuration of the device produces flickering or blanking of the LCD display.

The serial monitor access from PC for configuration of the device requires USB-to-Serail adaptors on pins PA9 and PA10, which couldn't be demonstrated in proteus due to it's limited USB functionalities.

These limitations remain mostly due to the time spent in solving simulation environment setup problems in proteus along with the tight submission deadline of the project.